SIMULATION ANALYTICAL REPORT

Strategy Profile: FIFO

Isolated Performance and Tail-Risk Analytics

GENERATED	MONTE CARLO RUNS	STRATEGIES TESTED
10 May 2026 13:56	**30**	**2**
METHOD	**UNIT**	**REPORT TYPE**
Discrete-Event Simulation (DES)	**Haemodialysis Centre**	**Comparative Performance Analysis**

Abstract

This report presents the results of a Monte Carlo discrete-event simulation study evaluating scheduling strategies for a haemodialysis unit. Stochastic variability in patient volume, machine availability, session duration, and nursing resources is modelled across 30 independent runs per strategy. Distributional summaries, tail-risk metrics (p90 / p99), and head-to-head strategy comparisons are provided to support operational decision-making.

Keywords: Monte Carlo | Discrete-Event Simulation | Haemodialysis | Scheduling | Tail-Risk

1. Simulation Setup

The parameters below fully specify the stochastic environment sampled during each Monte Carlo iteration. All stochastic quantities are drawn independently per run using a seeded Mersenne Twister PRNG.

TEMPORAL BOUNDS

Parameter	Value	Unit	Description
Shift Duration	360	*min*	Hard wall -- 6 h total shift
Nominal Session Duration	210 - 240	*min*	Prescribed per-patient session (~4 h target)
Min Viable Session	180	*min*	Sessions shorter than this are marked failed (3 h)
Machine Cooldown	60	*min*	Mandatory inter-session machine rest period

RESOURCE RANGES (uniform draw per iteration)

Parameter	Value	Unit	Description
Total Machines	15 - 20	*units*	Physical dialysis stations available
Patient Volume	15 - 20	*patients*	Demand sampled each shift
Nursing Staff	2 - 4	*nurses*	Concurrent supervision capacity
Session Duration	210 - 240	*min*	Per-patient dialysis time
Machine Ready Delay	0 - 90	*min*	Post-cooldown preparation lag

STOCHASTIC EVENT GENERATORS

Parameter	Value	Unit	Description
Patient Arrival Offset	Uniform[0, 60]	*min*	Arrival jitter relative to shift start
Setup Duration	Uniform[10, 20]	*min*	Machine preparation time per patient

PROBABILITY METRICS

Parameter	Value	Unit	Description
Machine Defect Probability	15.0%	--	Independent Bernoulli draw per machine per session

EXPERIMENT PARAMETERS

Parameter	Value	Unit	Description
Monte Carlo Iterations	30	*runs*	Independent replications per strategy

2. Statistical Summary

Distributional metrics across all Monte Carlo iterations for this strategy.

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	28.8 +/- 12.3	28.2	47.1	51.0
Max Wait Time (min)	52.6 +/- 19.2	49.0	80.2	91.6
Avg Session Time (min)	224.9 +/- 2.5	224.8	227.3	230.7

3. Distribution & Utilisation

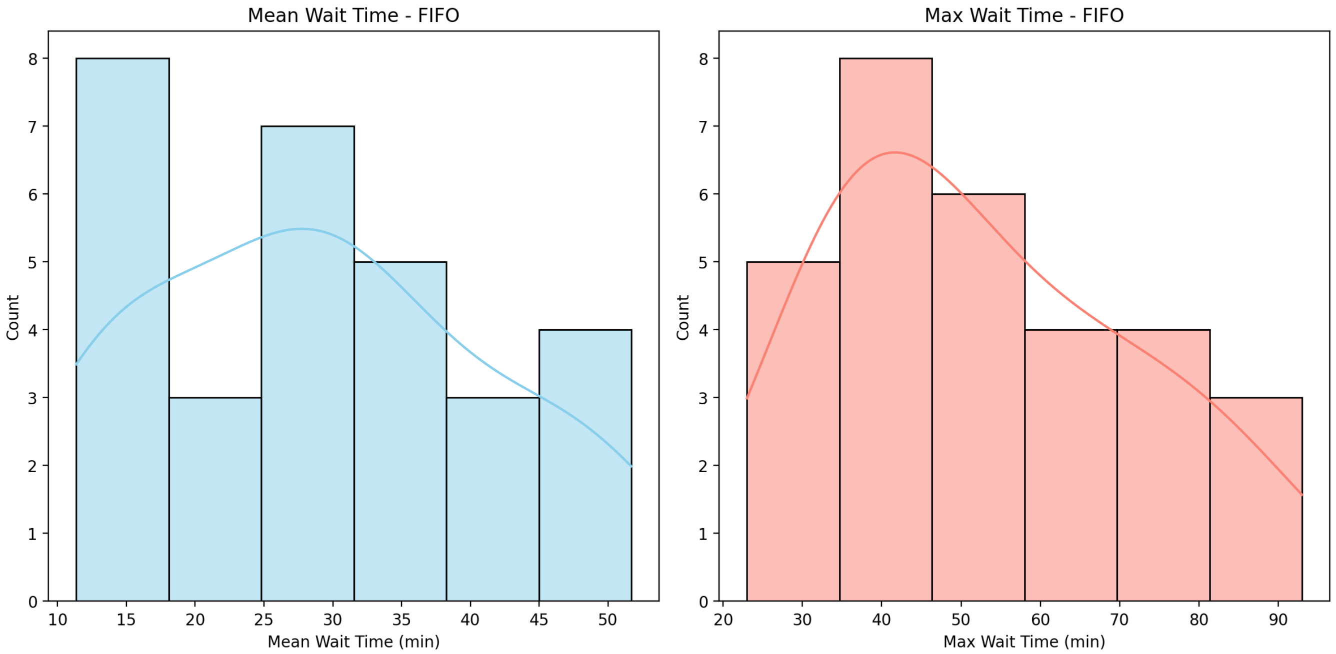

Figure 1. Wait-time distribution across Monte Carlo runs.

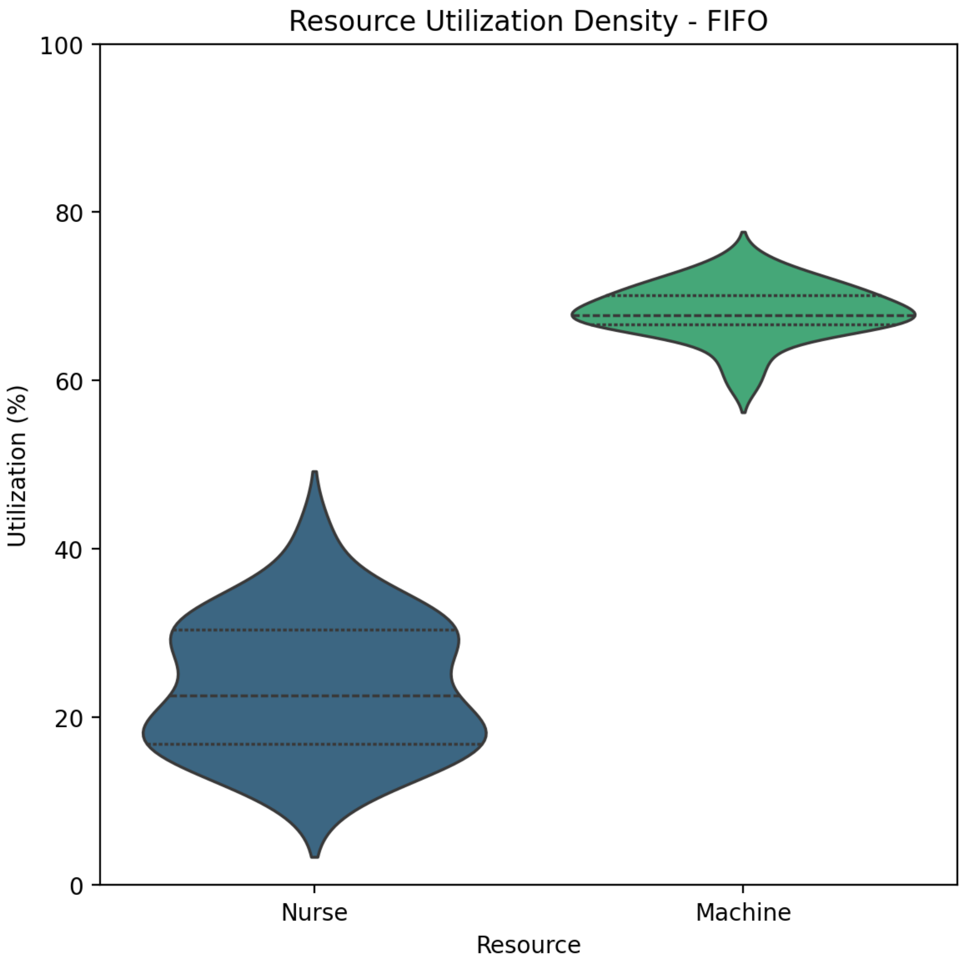

Figure 2. Machine and nurse utilisation.

4. Advanced Insights

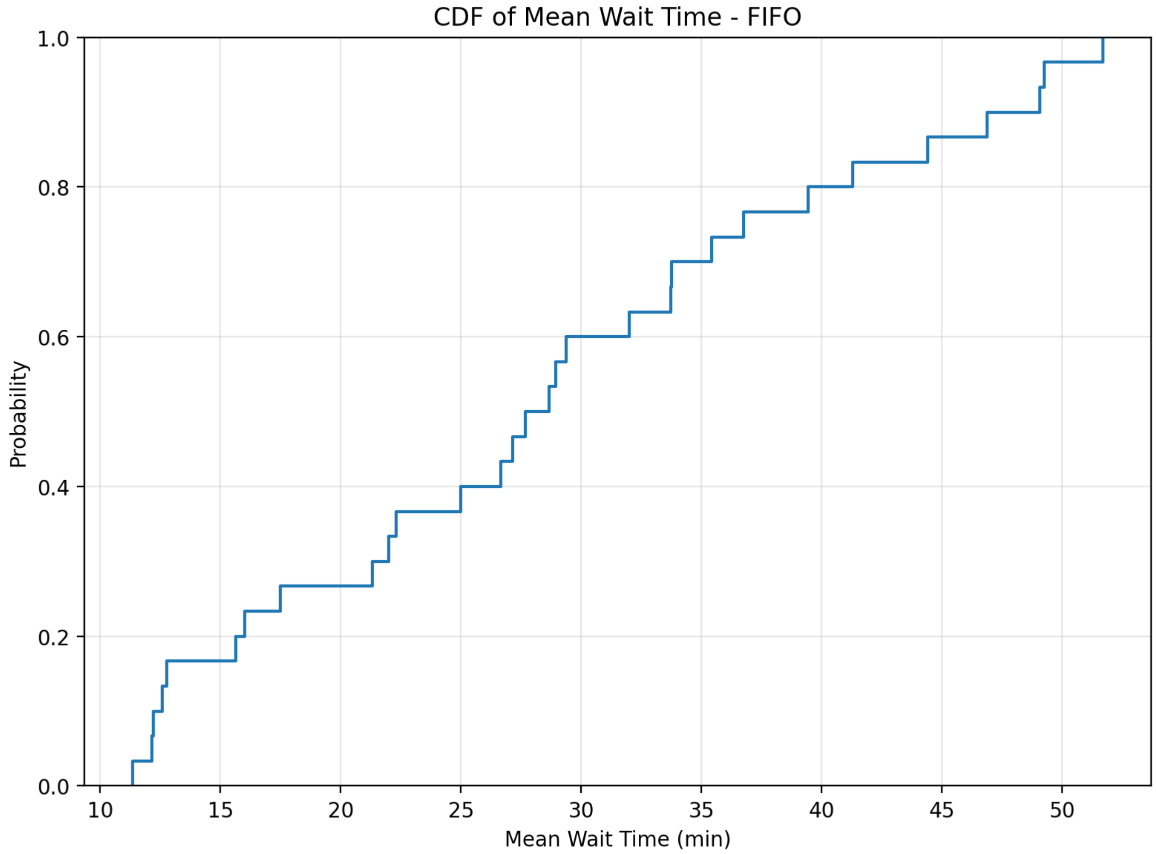

Figure 3. Empirical CDF of mean wait time.

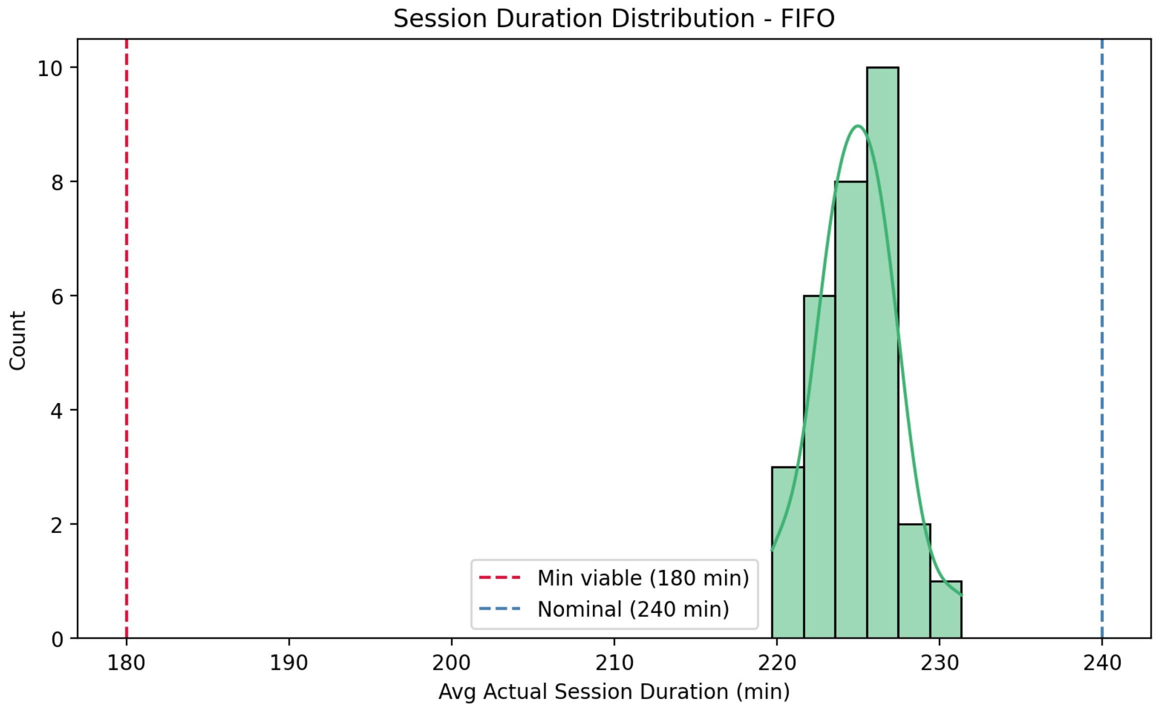

Figure 4. Actual session duration distribution (3 h min / 4 h nominal reference lines).

5. Edge-Case Stress Tests

Performance under extreme, deterministic boundary conditions designed to expose failure modes not visible in the central Monte Carlo distribution.

Scenario: Burst Arrival

[FAIL]" if failed else "[PASS] Strategy: FIFO | Wait (mean / max): 66.5 / 139.0 min | Avg session: 220 min | Truncated: 4 | Failed patie

Scenario: Maintenance Disaster

[FAIL]" if failed else "[PASS] Strategy: FIFO | Wait (mean / max): 0.0 / 0.0 min | Avg session: 215 min | Truncated: 0 | Failed p

Scenario: Marathon Shift

[FAIL]" if failed else "[PASS] Strategy: FIFO | Wait (mean / max): 0.0 / 0.0 min | Avg session: 230 min | Truncated: 5 | Failed p